

Jinil Kim

haba6030@snu.ac.kr | Homepage | GitHub | LinkedIn

RESEARCH INTERESTS

- *Computational Neuroscience* (brain–behavior principles, shared latent structure)
- *Human-Inspired AI and AI for Neuroscience*
- *Clinical Translation* (personalized diagnosis and intervention)

EDUCATION

Seoul National University

B.S. Computer Science · B.A. Psychology · B.A. Linguistics

Seoul, South Korea

Mar 2021 – Exp. Feb 2027

GPA: 4.18 / 4.3 (3.97 / 4.0) · 9 semesters (Mandatory military service, Jul 2023 – Jan 2025)

The University of Hong Kong

Summer Research Program, Faculty of Psychology

Hong Kong SAR

Jun 2025 – Aug 2025

PUBLICATIONS & MANUSCRIPTS

Workshop Papers

- **Kim, J.**, Cho, A. M., Seo, J., & Cha, J. (2026). *Inferring individualized color-vision distortions from fMRI hue-representation geometry*. International Conference on Machine Learning (ICML) 2026 Workshop on Structured Data for Health, Seoul, Korea. [paper]

Under Review

- **Kim, J.**, Jeong, J., Yun, H., Oh, M.-H., & Ahn, W.-Y. (2026). *Inverse reinforcement learning of latent human planning*. Under review at the Annual Conference on Neural Information Processing Systems (NeurIPS).
- **Kim, J.**, & Ryu, J. (2026). *Conversational relevance in the conjunction fallacy is represented independently in language models*. Under review at the NeurIPS 2026 Workshop on Linguistic Principles for Foundation Models (LP4FM).

In Preparation

- **Kim, J.**, Cho, A. M., Seo, J., & Cha, J. *Preserved categorical identification with impaired hue interpolation in color vision deficiency motivates a cortically derived correction filter*.
- **Kim, J.**, & Yao, Y.-W. *Depression relates to when, and not only how strongly, information enters a social decision*.

CONFERENCE PRESENTATION

- **Kim, J.**, Cho, M., Seo, J., & Cha, J. (2026). *fMRI decoding reveals intact neural color representations in color vision deficiency*. OHBM 2026 Annual Meeting, Bordeaux, France. [Poster]

RESEARCH EXPERIENCES

Connectome Lab

Undergraduate RA (Advisor: Prof. Jiook Cha)

Seoul National University

Apr 2025 – Present

- Led a project characterizing distortions of cortical hue geometry in color vision deficiency from fMRI. Translated them into personalized filters that moved hue-discrimination thresholds to the control range.
- Proposed an uncertainty quantification pipeline for an EEG neural field foundation model, measuring whether predicted uncertainty matches actual error and using it to guide electrode placement.

Computational Clinical Science Lab

Undergraduate RA (Advisor: Prof. Woo-Young Ahn)

Seoul National University

Apr 2025 – Present

- Developed SC-AIRL, a cognitively informed inverse RL framework raising action prediction accuracy by 31% over standard AIRL. The inferred planning depth significantly correlated with trait anxiety.

MIND Lab

Visiting Researcher (Advisor: Prof. Yuan-Wei Yao)

University of Hong Kong

Jun 2025 – Aug 2025

- Adapted a time-varying drift-diffusion model to social decision data with an onset-latency parameter. Showed that the depressed-range group integrates the reward–emotion contrast 0.29 s later ($d = 0.32$).

SCHOLARSHIPS & HONORS

KOSAF Humanity 100 Years Undergraduate Scholarship	(\$12,500)	Mar 2025 – Jul 2026
SNU Undergraduate Research Grant	(\$3,500)	Apr 2025
Best Poster Award, SNU Undergraduate Research Program		Jan 2026
SNU Student-Proposed Liberal Arts Course Contest	(1st Place)	Jul 2023

RELEVANT COURSEWORK

Neuroscience	Neuroscience · Data Science in Human Neuroimaging · Clinical Neuropsychology · Cognitive Processes · Computational Modeling in Experimental Psychology · Brain, Mind, and Behavior
Computer Science	Data Structures · Algorithms · Computer Architecture · Logic Design · Computer Programming · Computational Linguistics
Mathematics & Statistics	Linear Algebra · Discrete Mathematics · Calculus I–II · Psychological Statistics · Advanced Psychological Statistics

ACADEMIC SERVICES

Conference Reviewer: NeurIPS 2026 Workshop on Foundation Models for the Brain and Body
Volunteer: ICML 2026

TEACHING EXPERIENCE

Basic Computing: First Adventure in Computing	Seoul National University
Teaching Assistant (Instructors: Yamada Akihiko, Heajung Min)	Autumn 2025 · Spring 2026
Instructed laboratory classes regarding Python and basic programming for local and international students; English-medium section in Spring 2026.	

LANGUAGES

Programming: Python (Advanced), R, C++, Java (Proficient), Stan (Intermediate)